Akoizo Original Practice - Form 3

Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Micas are representative…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Sheet silicates

B. Native elements

C. Halides

D. Sulfides

Answer: _____

2. (2 points) Angular gravel clasts suggest…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Breccia

B. Conglomerate specifically

C. Obsidian

D. Schist

Answer: _____

3. (2 points) A rock can contain…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Only one chemical element

B. No geological material

C. Only transparent crystals

D. Multiple minerals

Answer: _____

4. (2 points) The approved way to distinguish halite is…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Taste every sample

B. Heat an unknown

C. Assume all clear crystals are salt

D. Physical evidence without tasting

Answer: _____

5. (2 points) The original rock is the…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Streak

B. Ore grade

C. Unit cell

D. Protolith

Answer: _____

6. (2 points) Acicular describes…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. A hardness value

B. A chemical reaction

C. Needlelike habit

D. One unique species

Answer: _____

7. (2 points) Bowen’s series is…

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. A hardness scale

B. A river classification

C. A conditional crystallization model

D. Every rock’s mandatory sequence

Answer: _____

8. (2 points) Under uncertainty, a useful answer states…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Only a color

B. An unrelated process

C. Evidence and a plausible alternative

D. Unjustified certainty

Answer: _____

9. (4 points) Compare pyroxene and amphibole cleavage.

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

Answer: _____
_____
_____

10. (4 points) Differentiate erosion and deposition.

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

Answer: _____
_____
_____

11. (4 points) Distinguish face from cleavage.

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

Answer: _____
_____
_____

12. (4 points) Name a carbonate and oxide.

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

Answer: _____
_____
_____

13. (4 points) Why are index minerals conditional?

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

Answer: _____
_____
_____

14. (4 points) How can veins constrain timing?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

15. (4 points) Contrast glass and fine crystals.

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

Answer: _____
_____
_____

16. (4 points) What is a poor timed strategy?

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

Answer: _____
_____
_____

17. (4 points) Why can quartz and feldspar differ?

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

Answer: _____
_____
_____

18. (4 points) Why avoid a unique environment from one grain size?

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

Answer: _____
_____
_____